# Individual Documentation Of
# Disease Detection_AI-Chatbot

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Section: **D**

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# 1. Problem Description:

### What is the project about?

In modern society, accessibility to primary healthcare remains a critical challenge. When people feel sick, they often search their symptoms online or wait days to see a doctor. This project creates a simple desktop application or web based where users can search and check their symptoms to get an immediate, educated guess about what illness they might have, along with descriptions and safety precautions. My specific part of Machine Learning model is Decision Tree.

### Why is this a challenging problem?

- **Overlapping Symptoms:** Many completely different diseases share the exact same foundational symptoms (like a high fever, headache, or vomiting). The machine learning model must find subtle patterns to tell them apart.
- **Imperfect Human Inputs:** In real life, patients often overlook or forget to report certain symptoms. The system needs to be robust enough to make a stable and accurate prediction even when data is missing or incomplete.

# 2. Methodology:

### How the system works?

Our project will be followed by a Decision Tree Machine Learning approach.

**Selection of Model:** For my component of the project, a Decision Tree Classifier was selected. This algorithm is exceptionally well-suited for medical diagnosis because its hierarchical branching structure inherently mirrors the step-by-step exclusionary logic used by human physicians during triage. To make the model robust against real-world human error such as a user neglecting to mention a minor symptom or misinterpreting a bodily change an intentional 5% stochastic symptom dropout layer was embedded into the data pipeline. This random noise forces the tree to generalize over broader symptom groups rather than overfitting to rigid, flawless data sequences.

**Data Transformation:** The dataset's textual features were handled using a binary one-hot vectorization workflow. A master template of all unique symptoms was established. Individual patient data strings were then mapped across this template, translating variable-length arrays of words into uniform mathematical rows composed entirely of 1s (symptom present) and 0s (symptom absent).

**Strategy:** To preserve structural balance and guarantee fairness across dozens of medical conditions, the data matrix was partitioned using a stratified 80% training and 20% testing split. The predictive capabilities of the model were strictly audited through the generation of Global Accuracy, Macro-Precision, Macro-Recall, and a Macro-F1 score, ensuring that less frequent, critical conditions were afforded equal mathematical weight alongside everyday common illnesses.

## 3. Implementation Details:

**Data Preprocessing:** Rigid data hygiene protocols were enforced to eliminate formatting conflicts and anomalies embedded within the raw CSV files. Any trailing spaces were programmatically stripped, and multi-word phrases were joined using underscores to generate standard, predictable feature keys. Additionally, the script automatically isolates and deletes unaligned rows, trailing commas, or empty placeholder lines from the spreadsheet layout to prevent structural array deformation.

**Backend & Model:** The core logic was written natively in Python. Upon launching the execution environment, the program loads the primary diagnostic data tables, purges structural spreadsheet artifacts, applies the 5% random dropout noise, and constructs the Decision Tree branches based on an Entropy Information Gain splitting matrix.

## 4. Results & Analysis:

Overall Model Accuracy : **94.11%**
Precision (Macro) : **96.47%**
Recall (Macro) : **94.11%**
F1-Score (Macro) : **94.74%**

The results show that using an entropy-based decision tree along with symptom noise filtering creates a reliable and accurate classification model. The model achieved a macro-averaged F1-score of nearly 95%, which is impressive given that it classifies more than 40 different diseases.

A closer look at the results shows that the model performs very well on diseases with unique and easily recognizable symptoms, such as AIDS and Diabetes, achieving a perfect score of 1.0000. However, it occasionally confused diseases with similar symptoms, such as GERD and Gastroenteritis or some respiratory illnesses. Since these conditions share common symptoms like nausea, stomach pain and coughing, removing even a single symptom through the 5% noise layer sometimes led the model to predict the wrong disease. Despite these challenges, the overall performance remained strong and consistent.